

Mark schemes

Q1.

Marking Instructions	AO	Marks	Typical Solution
Uses sum of probs = 1	AO1.2	M1	$0.4 + b + c = 1$
Uses formula for $E(R)$	AO1.1a	M1	$b + c = 0.6$
Uses formula for variance $E(X^2) - (E(X))^2$	AO1.1a	M1	$E(R) = 0.2 \Rightarrow$ $(-2 \times 0.3) + (0 \times b) + (a \times c) + (4 \times 0.1) = 0.2$
Obtains a , b and c CAO	AO1.1b	A1	$ac = 0.4$ $E(X^2) - (E(X))^2 = (4 \times 0.3) + (0 \times b) + (a^2 \times c) + (16 \times 0.1) - (0.2)^2 = 3.56$ $a^2c = 0.8$ From (2) and (3) $a = 2$ Hence $c = 0.2$ and $b = 0.4$
Total 4 marks			

Q2.

(a) Mean = $0 \times 0.1 + 1 \times 0.35$
AG

M1

= 1.85

$E(X^2) = 0^2 \times 0.1 + 1^2 \times 0.35$...

Full method including - 1.85²

M1

= 4.75 $\text{Var}(X) = 4.75 - 1.85^2$

For $E(X^2) = 4.75$

A1

= 1.3275

For final answer AWRT 1.33

A1

(b) (i) $T = c + nX$

B1

1

(ii) $E(c + nX) = c + nE(X)$

Getting at least as far as $c + E(nX)$

M1

$= c + 1.85n$

CAO

A1

$\text{Var}(c + nX) = \text{Var}(c) + \text{Var}(nX)$

Getting at least as far as $(0 +) \text{Var}(nx)$

M1

$= 0 + n^2 \text{var}(X) = 1.3275n^2$

FT their $\text{Var}(X)$ if $0 < \text{Var}(X) < 4$

A1F

4

[9]

Q3.

(a) (i) $1 - \left(\frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6}\right)$

OE

M1

$= \frac{1}{20} = 0.05$

AG

A1

2

(ii) $E(X) =$

$1 \times \frac{1}{3} + 2 \times \frac{1}{4} + 3 \times \frac{1}{5} + 4 \times \frac{1}{6} + 5 \times \frac{1}{20}$

At least 2 terms

M1

$= 2.35$

OE: give B2 for only 2.35 seen

A1

2

(iii) $E(X^2) =$

$1 \times \frac{1}{3} + 4 \times \frac{1}{4} + 9 \times \frac{1}{5} + 16 \times \frac{1}{6} + 25 \times \frac{1}{20}$

(= 7.05)

All 5 terms

$E(X^2) = 7.05$ with no working scores M0

Correct working but labelled Var(X) and then no more done also scores M0

M1

$$\text{Var}(X) = E(X^2) - E(X)^2$$

Applied to this problem

m1

$$= 1.5275$$

AG

A1

3

(iv) $1 - \left(\frac{1}{3} + \frac{1}{4}\right)$ or $\left(\frac{1}{5} + \frac{1}{6} + \frac{1}{20}\right)$

M1

$$= \frac{5}{12} \text{ or } 0.417$$

AWRT Accept answer only for B2

A1

2

(b) $'2.35' \times 100 - 50$

Their value of mean

M1

$$= 185$$

FT from (a)(ii)

Give B2 for only 185 seen

A1F

$$100^2 \times 1.5275 \text{ or } 100 \times \sqrt{1.5275}$$

M1

$$\text{SD} = \sqrt{15275} = 5\sqrt{611} = 124$$

AWFW 123.5 to 124 or $5\sqrt{611}$

Give B2 for only 123.5 to 124 or $5\sqrt{611}$ seen

A1

4

[13]

Q4.

(a)

n	Outcome	$P(N = n)$
1	H	0.5 ($\frac{1}{2}$)
2	TH	0.25 ($\frac{1}{4}$)
3	TTH	0.125 ($\frac{1}{8}$)
4	TTTH	0.0625 ($\frac{1}{16}$)
5	TTTTA	0.0625 ($\frac{1}{16}$)

B1 for **one** correct entry for $n = 1, 2, 4$

B2 for all 3 correct

Can be implied by correct $E(N)$

B2,1

$$E(N) = \left(1 \times \frac{1}{2}\right) + \left(2 \times \frac{1}{4}\right) + \left(3 \times \frac{1}{8}\right) + \left(4 \times \frac{1}{16}\right) + \left(5 \times \frac{1}{16}\right)$$

$$\sum_{n=1}^{n=5} n \times P(N = n)$$

(all 5 terms attempted / seen / implied)

$$= \frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{4}{16} + \frac{5}{16} = \frac{31}{16} = 1\frac{15}{16} \text{ (1.9375)}$$

(awfw 1.93 to 1.94)

M1

A1

4

(b)

m	Outcome	$P(M = m)$
1	H	($\frac{1}{4}$)
2	TH	$\frac{3}{4} \times \frac{1}{4} = \frac{3}{16}$
3	TTH	$\left(\frac{3}{4}\right)^2 \times \frac{1}{4} = \frac{9}{64}$
4	TTTH	$\left(\frac{3}{4}\right)^3 \times \frac{1}{4} = \frac{27}{256}$
5	TTTTA	$\left(\frac{3}{4}\right)^4 \times 1 = \frac{81}{256}$

(given)

(given)

(B1 any one correct)

(B2 any 2 correct)

(B3 all 3 correct)

B3,2,1

3

(c) (i) $P(J, R)$:

$$P(1,1) = \frac{1}{2} \times \frac{1}{4} = \frac{1}{8} \text{ (oe)}$$

e.g 0.125

attempt at any $P(n,n)$

M1

$$P(2,2) = \frac{1}{4} \times \frac{3}{16} = \frac{3}{64} \quad (\text{oe})$$

$$P(3,3) = \frac{1}{8} \times \frac{9}{64} = \frac{9}{512} \quad (\text{oe})$$

$$P(4,4) = \frac{1}{16} \times \frac{27}{256} = \frac{27}{4096} \quad (\text{oe})$$

any 1 correct **to 3sf**

A1

$$P(5,5) = \frac{1}{16} \times \frac{81}{256} = \frac{81}{4096} \quad (\text{oe})$$

any 5 correct **to 3sf**

A1

$$p = \sum_{n=1}^{n=5} P(n,n)$$

$\sum_{n=1}^{n=5} P(n,n)$ with all 5 values attempted

m1

$$\Rightarrow p = \frac{221}{1024} \quad (0.2158)$$

(awfw 0.215 to 0.217)

A1

©2025 Exam Papers Practice. All rights reserved.

5

$$(ii) = 3 \times \left(\frac{221}{1024} \right)^2 \times \left(\frac{803}{1024} \right)$$

(either term with **their** p used)
 $(0 < p < 1)$

M1

$$+ \left(\frac{221}{1024} \right)^3$$

(Second term with **their** p used)
 $(0 < p < 1)$

M1

$$P(X \geq 2) = P(X = 2) + P(X = 3)$$

dep (M1M1)

Mdep1

$$= 0.120 \text{ (3dp)}$$

(allow 0.119; 0.12; 0.121)

A1

4

[16]

Q5.

(a) $\binom{n}{m} p^m (1-p)^{n-m} \geq \binom{n}{m-1} p^{m-1} (1-p)^{n-m+1}$

Stated or used

M1

$$\frac{n!}{m!(n-m)!} p^{n-m} \geq \frac{n!}{(m-1)!(n-m+1)!} (1-p)^{n-m+1}$$

Expanding $\binom{n}{m}$ and $\binom{n}{m-1}$

m1

$$(n-m+1)p \geq m(1-p)$$

$$np - mp + p \geq m - mp$$

Simplification

A1

$$m \leq (n+1)p$$

Fully correct and complete derivation

AG

©2025 Exam Papers Practice. All rights reserved.

A1

4

(b) B(10, 0.65)

$$(n+1)p = 7.15 \text{ and } (n+1)p - 1 = 6.15$$

Can be implied by '7.7'

M1

$$m = 7$$

CAO

A1

B(35, 0.5)

$$(n+1)p = 18 \text{ and } (n+1)p - 1 = 17$$

(M1)

$m = 17 \text{ and } 18$

CAO; accept 'or'

A1

3

[7]

Q6.

(a) (i) $E(X) = 1.00 \times 0.35 + 1.50 \times 0.21 + 2.00 \times 0.24 + 3.00 \times 0.20 = £1.745$

M1

(ii) $E(X^2) = 1.00^2 \times 0.35 + 1.50^2 \times 0.21 + 2.00^2 \times 0.24 + 3.00^2 \times 0.20 =$

3.5825

3.58 ~ 3.59

M1

A1

(iii) standard deviation = $\sqrt{3.5825 - 1.745^2} =$
for variance (but not if called standard deviation)
for standard deviation

M1

£0.733

0.73 ~ 0.734

allow 0.537 (0.537 ~ 0.538) if called variance

M1 A1

6

(b) (i) $0.24 + 0.20 = 0.44$
 cao

M1 A1

(ii) mode = 1

B1

Probability $\geq$ mode = 1
 cao

B1

4

(c) (i) decrease –

B1

children under 5 are charged the least (zero).

B1

(ii) increase –

B1

spread of distribution increased.

B1

4

[14]

Q7.

(a) (i) $E(X) = 0 \times 0.22 + 1 \times 0.25 + 2 \times 0.2 + 3 \times 0.14 + 4 \times 0.19$
 $= 1.83$

method

M1

1.83 *cao*

A1

(ii) $E(X^2) = 0 \times 0.22 + 1 \times 0.25 + 4 \times 0.2 + 9 \times 0.14 + 16 \times 0.19$
method

M1

$= 5.35$

(iii) $V(X) = 5.35 - 1.83^2 = 2.001$
method for variance – may be calculated as $E[(X - E(X))^2]$
disallow if called standard deviation

M1

s.d. = $\sqrt{2.0011}$
variance

m1

$= 1.41$

1.41 - 1.42

allow for variance = 2.00 (2.00 - 2.01)

A1

6

(b) (i) $P(X < 3) = 0.22 + 0.25 + 0.2 = 0.67$

0.67 cao

B1

- (ii) $P(X > 1.83) = 0.2 + 0.14 + 0.19$
'their' mean
requires correct method for mean

m1

$$= 0.53$$

0.53 cao

A1

- (iii) median = 2
 2 cao

B1

$$P(X > 2) = 0.14 + 0.19 = 0.33$$

0.33 cao

B1

5

[11]

EXAM PAPERS PRACTICE

©2025 Exam Papers Practice. All rights reserved.

Examiner reports

Q2.

Part (a) was almost universally answered correctly. Most students then went on to complete the question successfully, although some found the algebra required confusing and squared the wrong part of the variance. Others ignored the instruction in part (b)(ii) to use the results from part (a) and so inevitably lost marks.

Q3.

The majority of students scored very well on this question, although a few treated X as a continuous variable and performed various integrations and so scored few, if any, marks. Students should have appreciated that, in part (a)(iii), where they were required to 'Show that $\text{Var}(X) = 1.5275$ ', simply quoting values such as $E(X^2) = 7.05$ without any working, was not acceptable. This 'general principal' can be expected to apply on future papers. Part (b) was well done, although a small proportion of students, having found the variance, did not proceed to the standard deviation.

Q4.

Most candidates were successful in completing the given table in part (a). Unfortunately, some then failed to calculate the value of $E(N)$ as requested whilst others changed the 'given' values in Table 1, thus losing marks. In part (b), Table 2 was often completed but not always successfully. However, candidates in the main then realised that they needed to use the values found in Table 1 and Table 2 to help them to calculate the required probabilities in part (c)(i). This usually resulted in at least 3 marks being gained. Their result to part (c)(i) was then often correctly used in part (c)(ii), again with at least 3 marks being gained.

Q5.

This final question proved very challenging to all but the best students. Correct, or even sensible attempts, at part (b) were reserved for the best students often due to an apparent lack of algebraic knowledge and related manipulative skill. This did not prevent such students from scoring at least 2, and often 3, marks in answering part (b).

Q6.

There was a good spread of marks on this question. Apart from candidates who confused variance with standard deviation or with $E(X)^2$, most were successful with part (a). In part (b) a few incorrectly identified the mode as 0.35. The most common error in part (c) was to say that since there was no charge for under 5s, the mean and standard deviation would be unchanged.

Q7.

This question was well answered. A few candidates thought that $E(X^2)$ and the variance of X were identical. Others confused variance and standard deviation. A number correctly identified and evaluated the variance but did not complete the calculation of the standard deviation. Some candidates were unable to identify the median and so unable to make progress in part (b)(iii). A few candidates used a normal distribution in part (b).